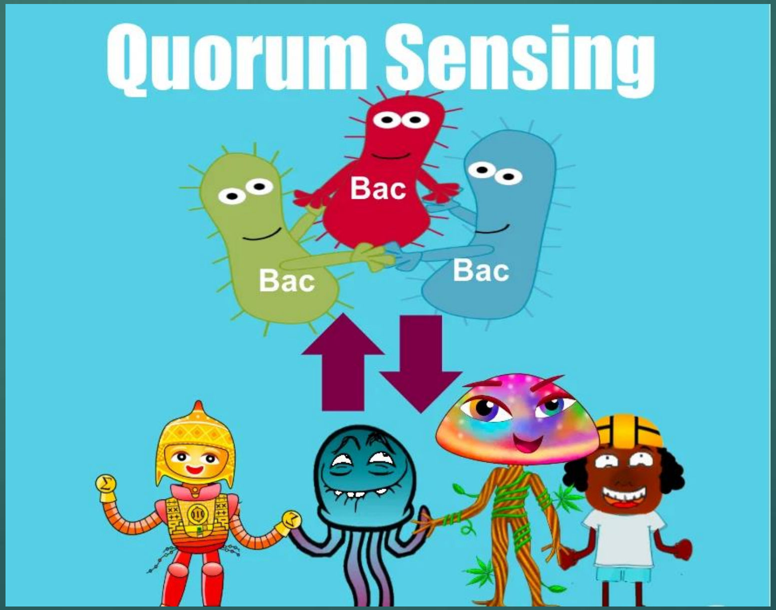
Cell Biology

Quorum Sensing



Dr. Dalia Mohsen Hamed

Program/Course Learning Outcomes:

Program Outcomes (NARS 2017):

Demonstrate understanding of knowledge of pharmaceutical, biomedical, social, behavioral, administrative, and clinical sciences.

Course Learning Outcomes:

a2: Identify the structure of prokaryotic and eukaryotic cells, process of gene expression and cellular communication.

a3: Recognize information about genetic codes in transcription , translation and mutation

c1: Interpret principles of cell structures, cell signaling, gene regulation, protein synthesis and mutation in treatment of disease.

Lecture outlines

- Quorum sensing definition.
- Quorum sensing (QS): a mode of microbial communication.
- Mechanism of quorum sensing.
- Signaling molecules involved in quorum sensing.
- The mechanism for AHL-based QS.
- The mechanism for AIP-based QS.
- Key Stages of Biofilm Formation.
- Biofilm as a complex surface-attached bacterial community.

Quorum Sensing

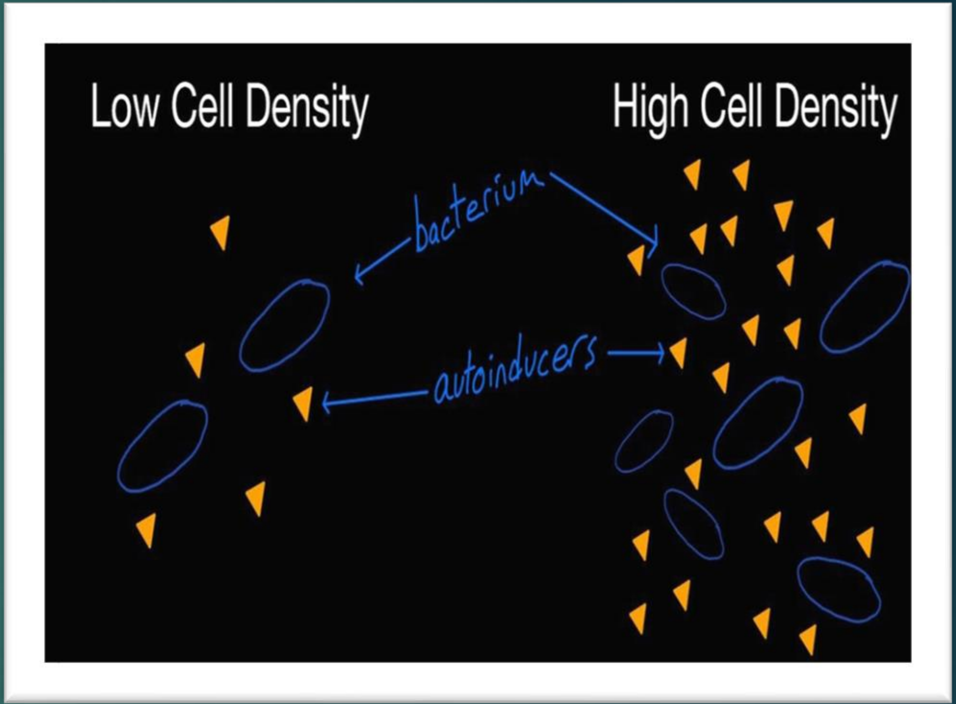
It is a process of cell-to-cell communication that allows bacteria to coordinate their behavior based on population density.

- Bacteria produce and detect small signaling molecules called autoinducers.

- As the bacterial population grows, the concentration of these molecules in the environment increases.

- Once a critical threshold (a quorum) is reached, it triggers a widespread change in gene expression across the entire population.

- This allows the group to act in a synchronized manner, essentially behaving like a multicellular organism.

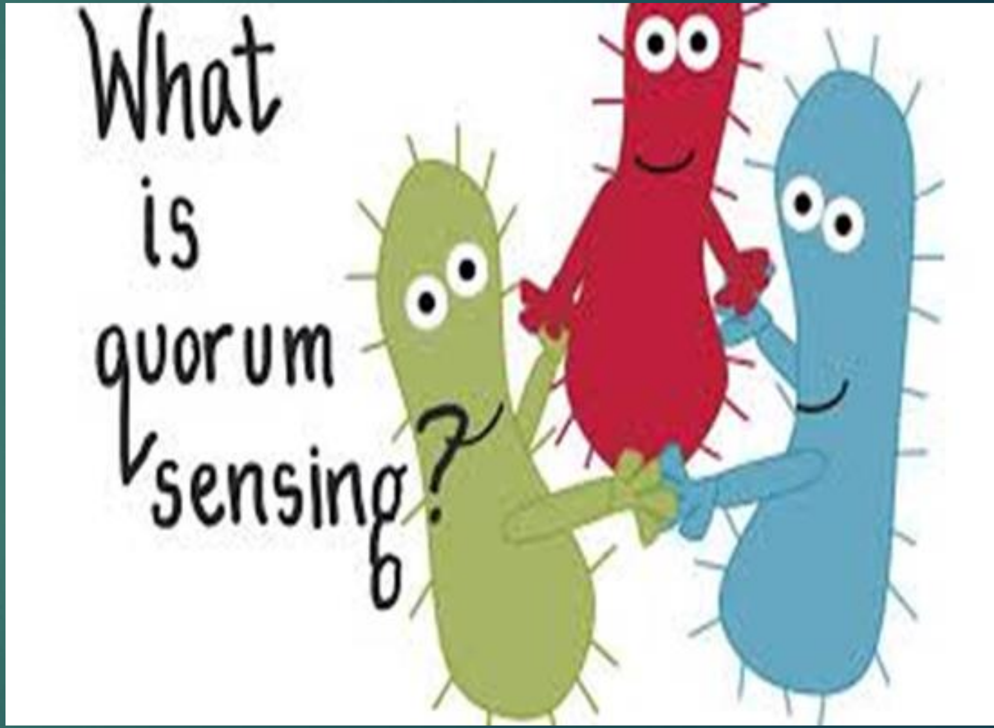


Quorum sensing (QS): a mode of microbial communication

- Cell-cell signaling systems.

- Ability to detect and respond to cell population density by gene regulation.

- Bacteria use QS to coordinate gene expression according to the density of their local population.

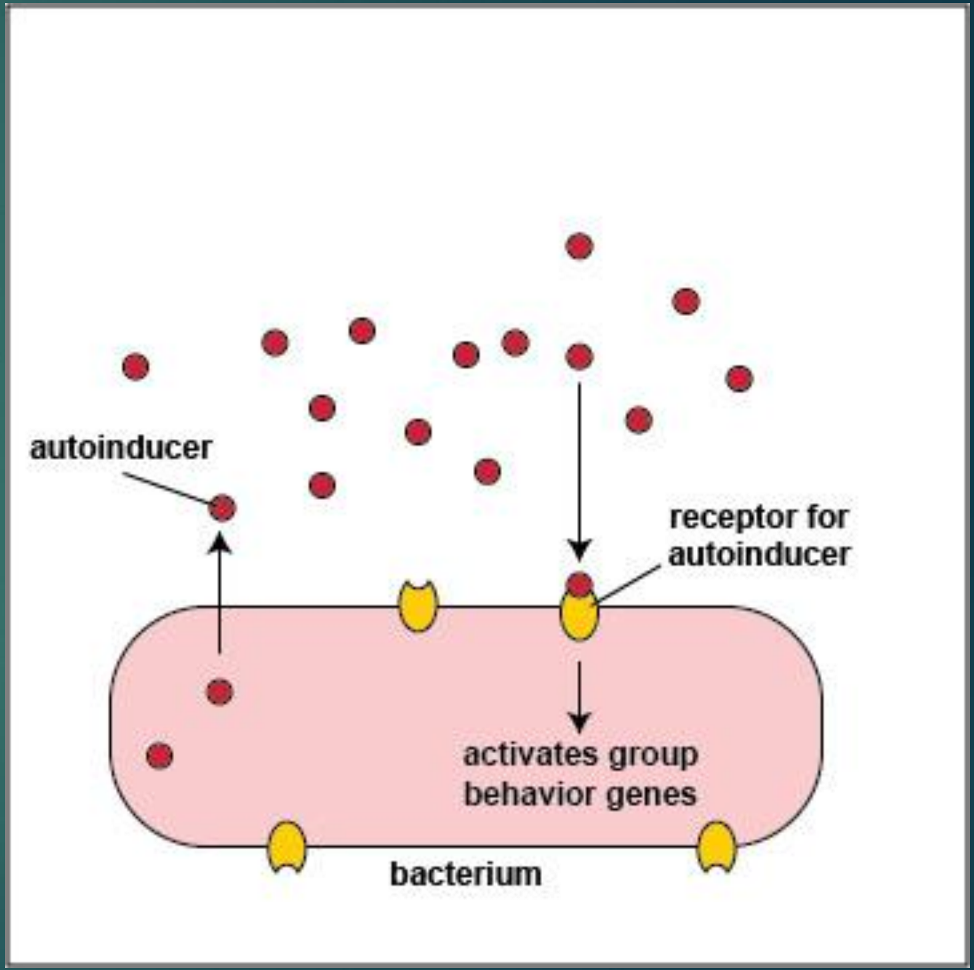


Mechanism of quorum sensing

1. Signal Production: Individual bacteria constantly produce and secrete low levels of **autoinducer molecules**.

2. Signal Accumulation: As the bacterial population grows, the extracellular concentration of these molecules increases proportionally.

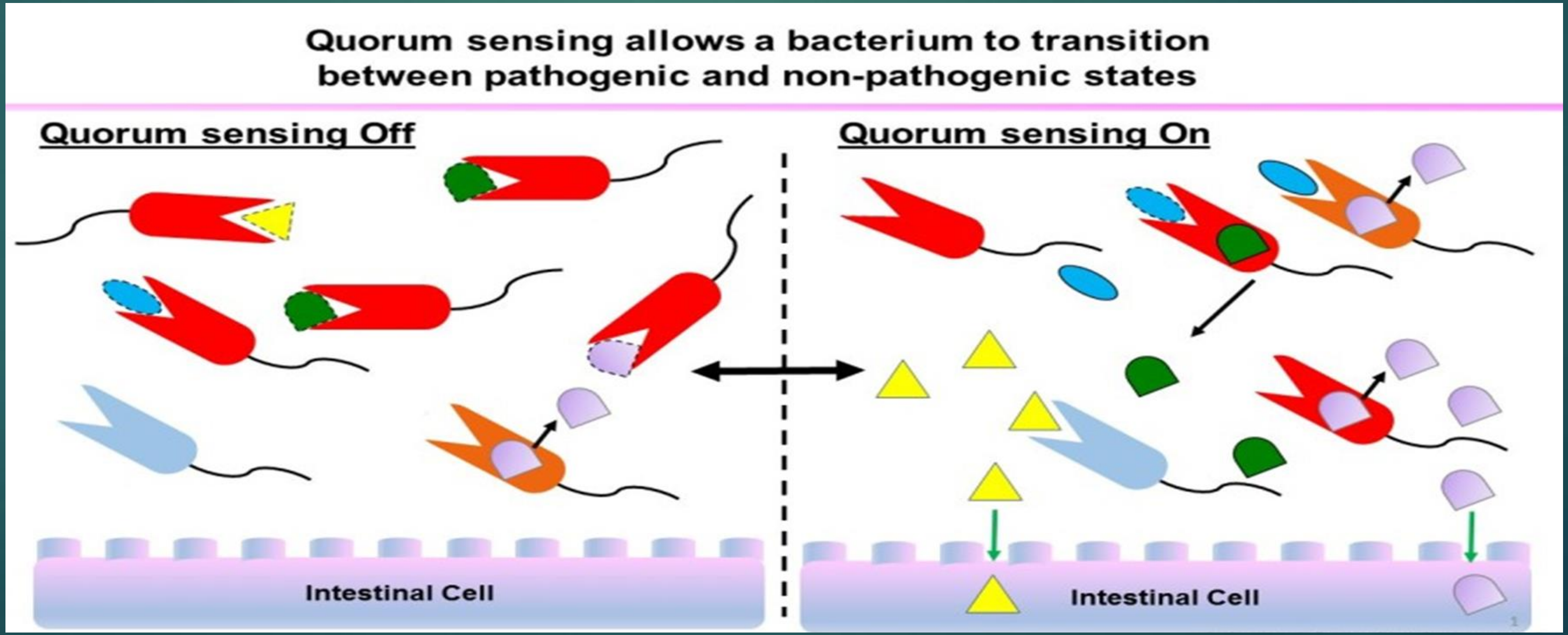
3.Signal Detection: When the autoinducer concentration reaches a critical threshold (a **quorum**), it binds to a specific receptor protein on or inside the bacterial cells.



Mechanism of quorum sensing (Cont.)

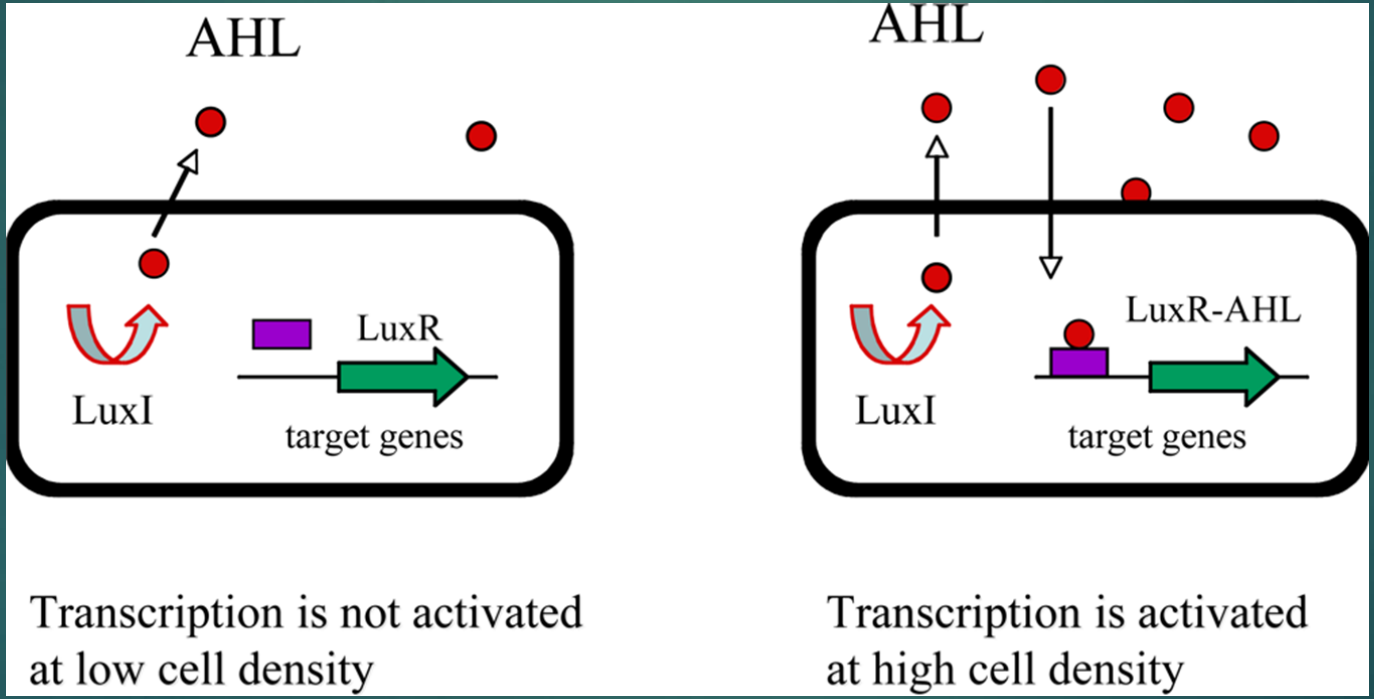
4.Gene Regulation: This binding activates the receptor, which then acts as a **transcription factor** to switch on (or sometimes off) specific sets of genes. The system **turns "on"** (activating gene expression for behaviors like virulence or bioluminescence) when signal concentration reaches a critical threshold, and remains **"off"** (low-level, basal expression) at low population density.

5. Synchronized Response: The entire population simultaneously alters its behavior.



Signaling molecules involved in quorum sensing

- **Acyl-Homoserine Lactones (AHLs):** Used primarily by **Gram- negative bacteria**.
- They diffuse across the cell membrane.



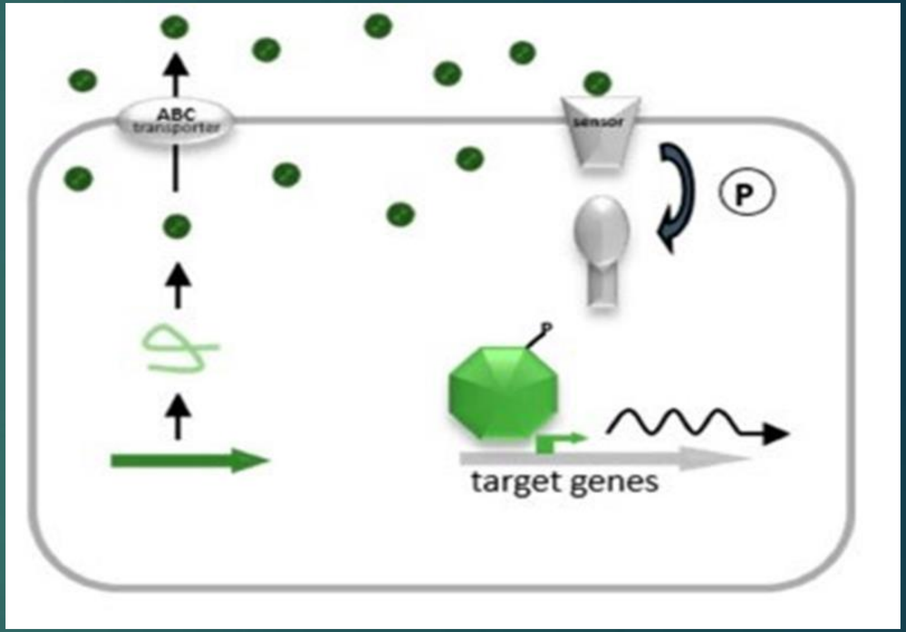
Quorum sensing Cell communication

https://www.youtube.com/watch?v=tjYUSpFw0lw

Signaling molecules involved in quorum sensing (Cont.)

- **Autoinducing Peptides (AIPs):**

Used primarily by **Gram-positive bacteria.** They are short peptides and are actively transported.



- **Autoinducer-2 (AI-2):** A "universal" signal molecule. Many different bacteria, both Gram-positive and Gram-negative, can produce and sense it, suggesting it allows for interspecies communication.

- **Others:** Specific systems like the *Pseudomonas* quinolone signal (PQS).

The mechanism for AHL-based QS:

LuxI-type synthase and a LuxR-type receptor.

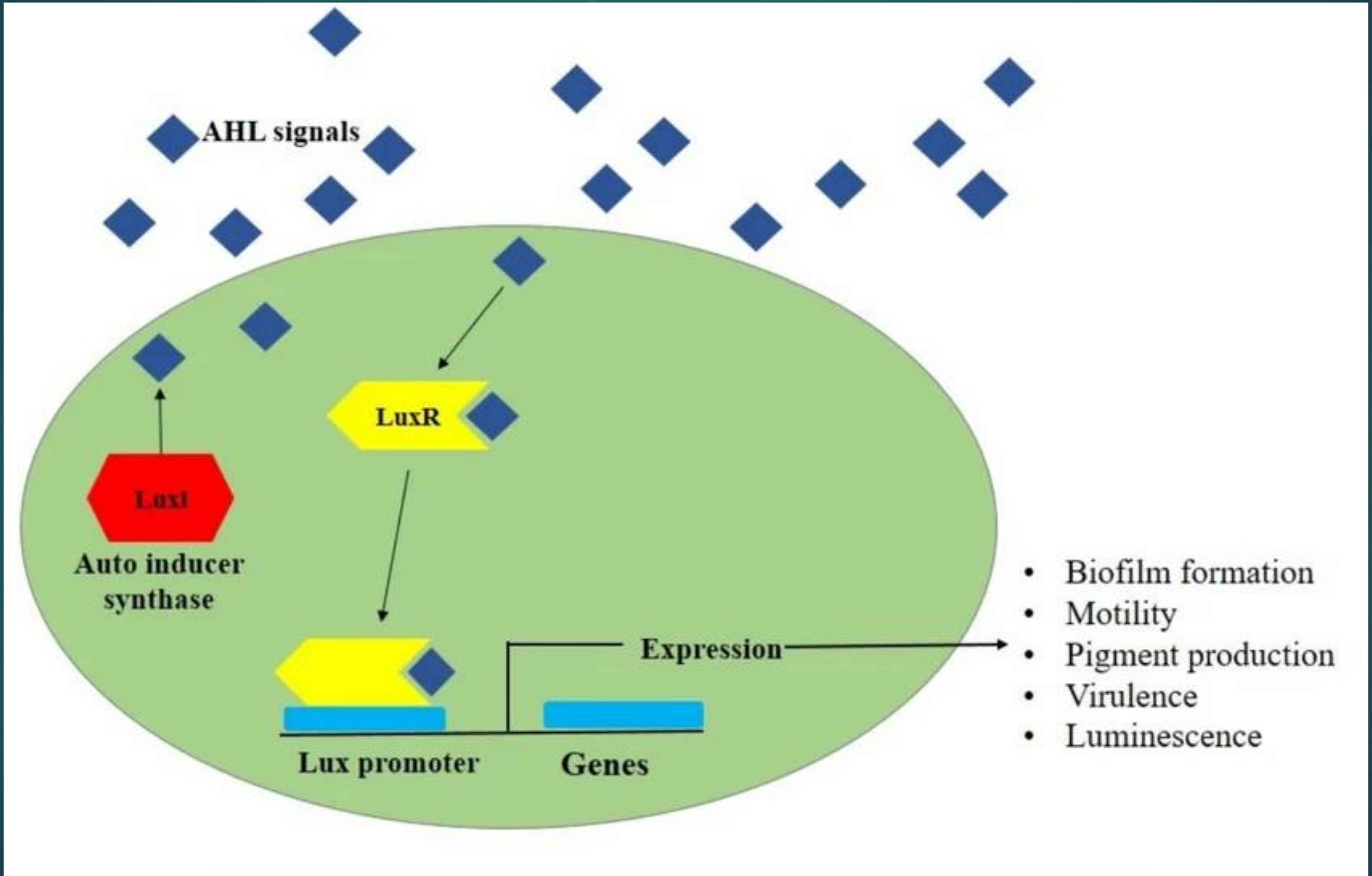
Step 1: Synthesis & Diffusion: AHLs are synthesized **at a low levels** via **LuxI-type synthases at** low cell densities and can diffuse out of the cell and into the environment .

Step 2: Accumulation: As the bacterial population grows, the extracellular concentration of AHL increases.

Step 3: Detection & Gene Activation: At a threshold concentration, the AHL binds to its **cognate LuxR-type transcriptional regulator** inside the cell. This **AHL-LuxR complex** then binds to specific promoter sequences on the DNA activating the transcription of target genes.

Step 4: Autoinduction (Positive Feedback): the target genes almost always include the gene for the **LuxI-type synthase** itself. This creates a powerful positive feedback loop: a little signal leads to more signal production, ensuring a rapid, population-wide, and synchronized response.

The mechanism for AHL-based QS:



The mechanism for AIP-based QS:

Autoinducing Peptides (AIPs) are the primary signaling molecules for quorum sensing in many Gram-positive bacteria, and a Two-Component System (TCS) is a common mechanism for detecting them, which consists of a transmembrane histidine kinase receptor and an intracellular response regulator.

How does it work?

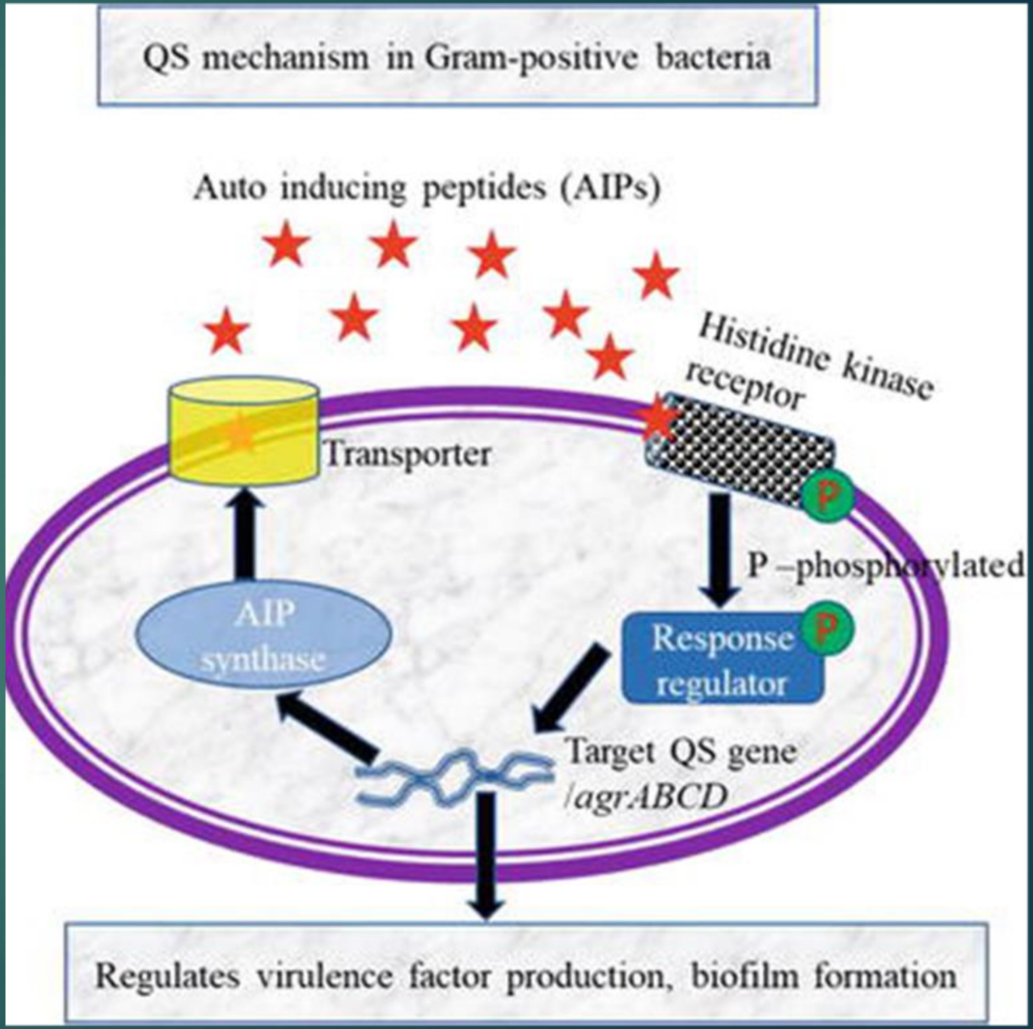
- **Signal Production & Release: Bacteria produce a precursor peptide, which is processed and secreted as the mature Autoinducing Peptide (AIP).**
- **Signal Detection: At a high population density, the extracellular concentration of AIPs rises. The AIP is detected by a membrane- bound sensor Histidine Kinase (HK).**

The mechanism for AIP-based QS (Cont.):

How does it work (Cont.)?

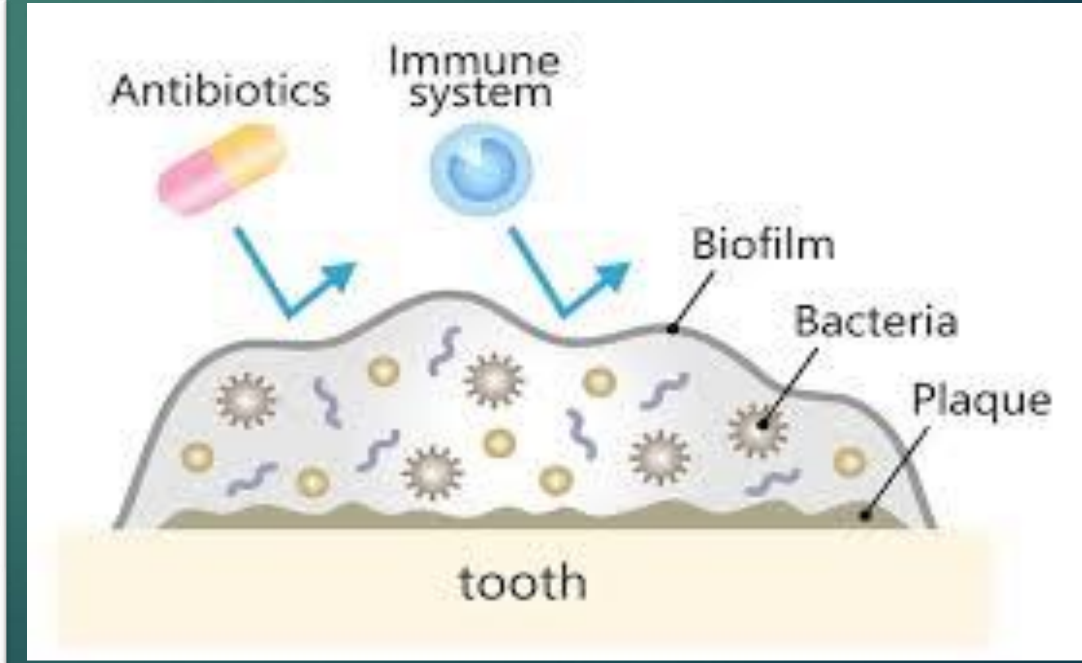
- **Phosphorylation Cascade:** The sensor HK autophosphorylates when it binds the AIP. It then transfers this phosphate group to a specific intracellular Response Regulator (RR) protein.

- **Gene Activation:** The phosphorylated Response Regulator undergoes a conformational change, allowing it to bind to DNA. This activates the transcription of target genes, such as those for virulence factors, biofilm formation, or toxin production



What is biofilm?

Biofilms are structured, slimy communities of microorganisms (bacteria, fungi, or protists) that adhere to surfaces—living or non-living—and are encased in a self-produced matrix of extracellular polymeric substances (EPS). These communities enable microbes to resist antibiotics, survive environmental stress, and share nutrients.



Key Stages of Biofilm Formation:

1. **Initial Attachment (Reversible):** Free-floating (planktonic) bacteria encounter a surface and bind loosely, often facilitated by flagella or electrostatic forces.

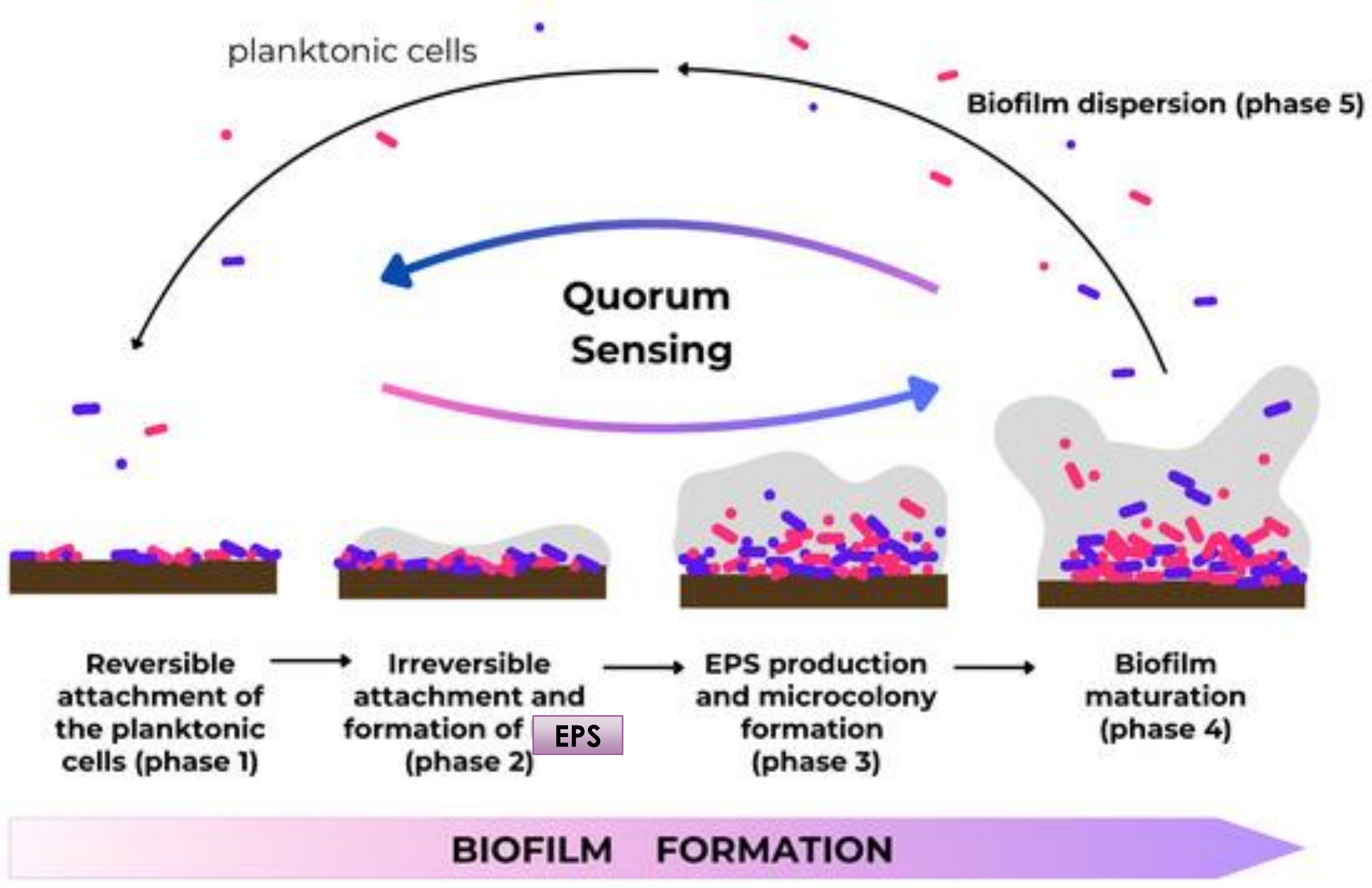
2. **Irreversible Attachment:** The bacteria produce an extracellular polymeric substances (EPS), forming a matrix that cements them to the surface, making them difficult to remove.

3. **Maturation I (Microcolony Formation):** The attached cells divide and cluster together, forming microcolonies and strengthening the EPS matrix.

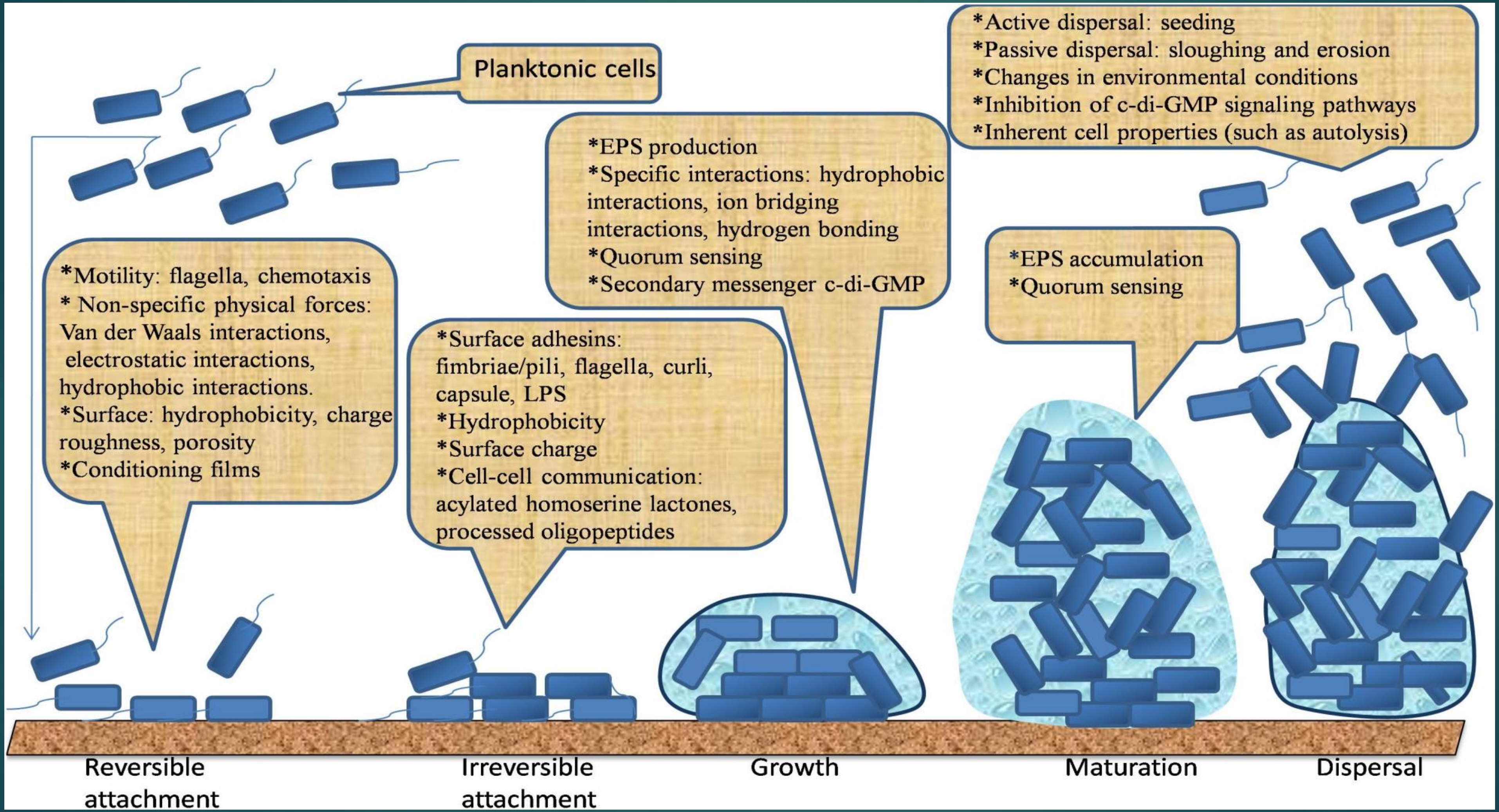
4. **Maturation II (Biofilm Architecture):** The biofilm grows into a complex structure, often with water channels.

5. **Dispersion:** Cells are released from the biofilm (often due to nutrient depletion) to return to a planktonic state and colonize new locations

Biofilm formation stages figure



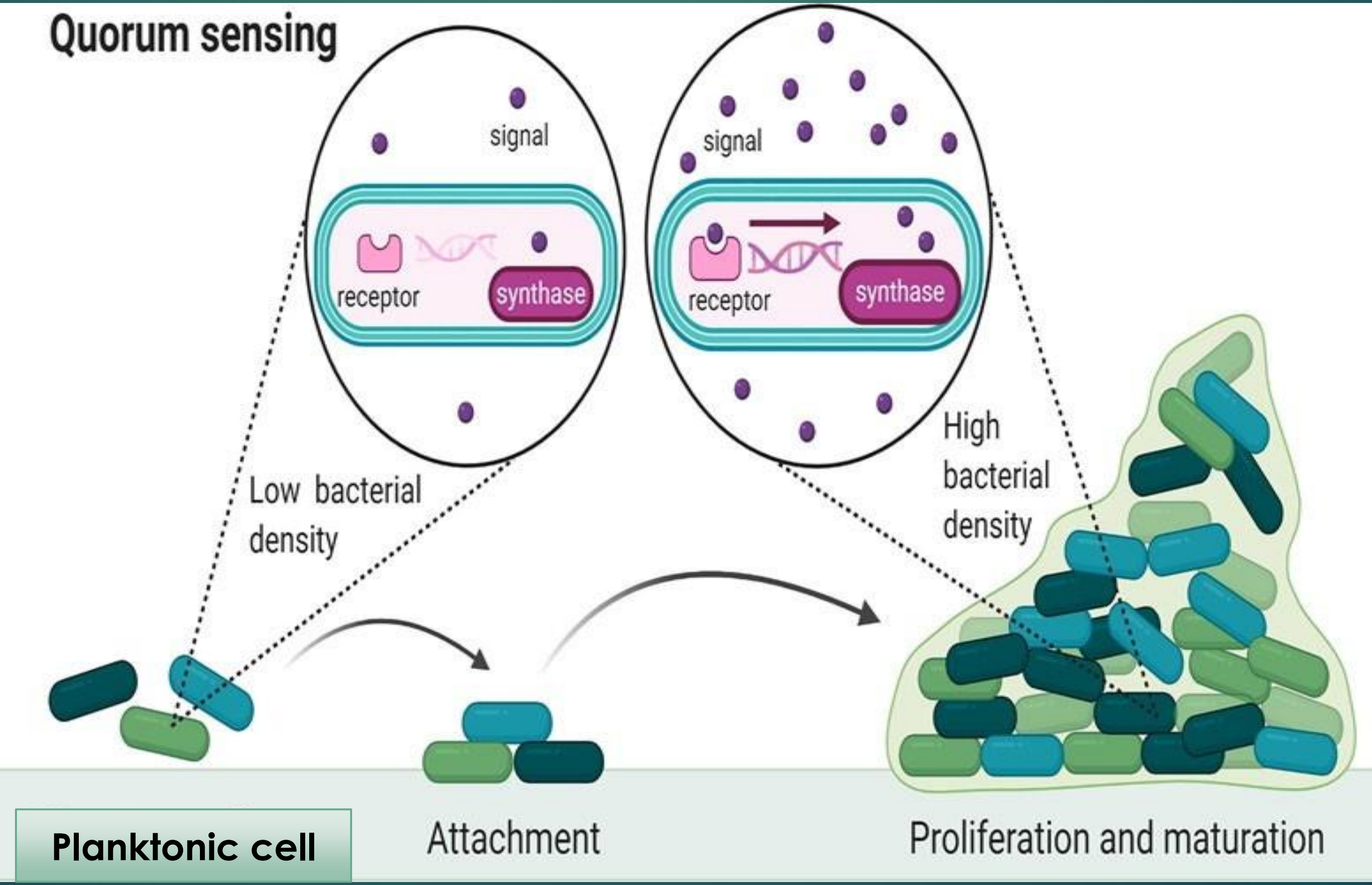
Biofilm as a complex surface-attached bacterial community



Bacterial Biofilm Formation Phases

Biofilm in healthcare and industries

Development of beneficial biofilms can be promoted through manipulation of adhesion surfaces, QS and environmental conditions.




Thank you